

Come Forth Into the Light of Things



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Come forth into the light of things, let nature be your teacher,” wrote William Wordsworth in his poem from 1798 titled *The Tables Turned*. Any conspiracy theorist worth his salt would have loved to jump on that line as a prediction of what is to come.

Today we see the Internet of Things weaving its magic to revolutionise lighting. The latest bulbs let you adjust the intensity and colour to mimic natural light. These can transform from a bright white-yellow like the morning Sun to a warmer and more golden-red part of the spectrum in the afternoon. Or these can allow you to simulate an entire golden forest, which is what Japanese electronics component manufacturer ROHM did just last year. What is enabling all this?

What happened to hardware

“A lot of changes have occurred since last year, such as the change from emitter technology to chip on board technology, different soldering processes for different chip manufacturers, driver requirements for various wattages and much more. This is a continuous process and a manufacturer needs to adapt according to

this dynamic change,” say Siva Satheesh, managing director, and Abdul Basith, executive director, Euro Lights, India.

Belkin WeMo devices run on Qualcomm’s Atheros AR9331 system on chip (SoC). Disney Research published a white paper where they discuss an innovative LED-to-LED communication system that uses the same SoC along with an 8-bit ATmega328p running Visible Light Communication (VLC) firmware and connected to an amplification board. It uses visible light to send data at a rate of up to 1kbps. The do-it-yourself bulbs are designed to interact with other gadgets that may not have full Wi-Fi connectivity and instead read data using VLC that uses visible light between 400THz and 800THz.

NXP’s GreenChip iSSL and GreenChip iCFL bring together wireless connectivity, energy-efficient lighting and low-power standby in a solution that supports software stacks like Zigbee and JenNet. JenNet-IP is a 6LoWPAN mesh-under-tree network with low memory footprint, specifically targeting low-power IEEE 802.15.4 based networking for residential and industrial applications. ZigBee LightLink is a standards based solution targeting in-room lighting for residential applications.

LEDSense is a new low-voltage driver architecture created by a company named Terralux. The company claims that they program LM-80 curve (an industry standard that lets users evaluate LEDs) into each product and monitor the temperature to ensure the LED is not overdriven for ambient temperature, and also ensure that their products comply with NEMA SSL6/7 dimming standards.

Back in India, OMRONICS Twilight Switch is a sunlight based automatic on/off sensor that switches on at dusk and switches off at dawn. “Improved dimming is possible by Android devices, infrared sensors and Bluetooth controllers these days. You also see motion sensors based

ROHM Luminary Promenade

